TEXORA · LECTURE SHEET

Nano Finishing and Nanocoating

Putting the nanoscale to work on finished fabric: the self-cleaning, antimicrobial and UV-protective finishes, and the three coating technologies the syllabus names — plasma polymerisation, layer-by-layer self-assembly and sol–gel coating.

TE-4206 · TECHNICAL TEXTILE AND NANO-TECHNOLOGY · TOPIC 9 · LEVEL 4 TERM 2

1. Nano finishing — what makes a finish “nano”?

A **nano finish** delivers its function through nanoscale material — particles, platelets or a nanometre-thin layer — applied to a textile by finishing machinery. Two advantages follow from the scale. First, **efficiency**: the active surface per gram is enormous (Topic 6), so add-ons of grams per kilogram do the work of much heavier conventional finishes. Second, **invisibility**: a layer tens of nanometres thick does not change the fabric's handle, drape or colour the way a thick coating does — the fabric stays a fabric. The governing weakness is the one named throughout this course: **durability** — what washing and abrasion do to a layer only atoms thick decides whether the finish is a product or a demonstration. Every finish below is therefore presented as function + mechanism + durability question.

2. The three headline finishes

2.1 Self-cleaning

Two different mechanisms share this name, and they must not be confused:

- **Photocatalytic cleaning (TiO_2).** UV light excites nano-titania (anatase), generating reactive species that oxidise organic soil and odorous compounds toward CO_2 and water; the surface simultaneously becomes highly water-loving, so rain sheets across it and lifts loosened dirt. Needs UV — weak indoors — and the same radicals can attack binder and fibre, so formulation balances activity against self-destruction.
- **Lotus-effect repellency.** A nano-rough hydrophobic surface (silica nanoparticles under a low-energy treatment) makes water bead into rolling droplets that pick up dirt as they roll. Nothing is chemically decomposed; the surface simply refuses to hold soil or water. Fails as the nano-roughness abrades away — durability is again the test.

2.2 Antimicrobial

Nano silver is the benchmark: Ag^+ ions released from the particle surface disrupt bacterial membranes and enzymes. Zinc oxide and MgO offer milder antimicrobial action combined with other functions. Design points: release must be fast enough to work and slow enough to last; the finish must be tested on the *washed* article; and antimicrobial claims are regulated — a laboratory kill rate does not license medical claims (the discipline of the YM-4203 sheet applies in full).

2.3 UV protective

Nano TiO_2 and ZnO absorb and scatter ultraviolet radiation across UV-A and UV-B. At nanoscale the particles do not whiten the fabric the way pigment-grade TiO_2 does — protection without a chalky look, which is precisely why the nano form is chosen. The product measure is UPF (ultraviolet protection factor): as a first estimate, a fabric transmitting a fraction T of UV has UPF ≈ 1/T, so 2% transmission corresponds to UPF 50. Durability and evenness decide the labelled rating, which must be measured on the finished, washed fabric.

FINISH	ACTIVE MATERIAL	MECHANISM	GOVERNING DURABILITY QUESTION
Self-cleaning (photo)	Nano TiO_2 (anatase)	UV-driven oxidation of soil; hydrophilic sheeting	Does activity survive washing — and does the catalyst attack its own binder?
Self-cleaning (lotus)	SiO_2 nanoparticles + hydrophobic agent	Nano-roughness → rolling droplets carry dirt away	How fast does abrasion remove the nano-roughness?
Antimicrobial	Ag, ZnO, MgO nanoparticles	Ion release; membrane/enzyme disruption	Release rate after repeated washing vs claim
UV protective	TiO_2, ZnO nanoparticles	UV absorption and scattering	UPF after washing and stretching

Table 1 — The syllabus's three finish families (self-cleaning counted twice, once per mechanism). "After washing" is part of every honest specification.

3. Nanocoating technologies

3.1 Plasma polymerisation

A low-pressure (or atmospheric) plasma — an ionised gas — is struck in a monomer vapour; the activated species deposit as an ultra-thin, highly crosslinked polymer film, nanometres to tens of

nanometres thick, on every exposed surface of the fabric. Because the film grows from the gas phase, it is conformal (it follows every fibre), uses no water or solvents, and can be tuned hydrophobic, hydrophilic or functional by monomer choice. Limits: vacuum equipment cost in the classic process, line-of-sight and batch character, and treating thick or bulky goods uniformly.

3.2 Layer-by-layer self-assembly (LbL)

The fabric is dipped alternately in solutions of oppositely charged polyelectrolytes (or charged nanoparticles); each dip deposits one charged molecular layer, which reverses the surface charge and invites the next. Thickness grows one nanometre-scale layer at a time and is counted in *bilayers* — five, ten, twenty dips, each adding a controlled increment. LbL builds flame-retardant intumescent stacks, antimicrobial layers and conductive assemblies with unmatched architectural control, at the price of process time: it is a dipping choreography, and industrial speed is its research frontier.

3.3 Sol–gel coating

A liquid precursor (commonly a silicon alkoxide such as TEOS) is hydrolysed to a *sol* of silica nanoparticles; the fabric is padded or dipped, and on drying and curing the sol gels into a thin, glassy SiO_2 network bonded to the fibre surface. The gel network is a host: it can carry TiO_2 for UV and self-cleaning, silver for antimicrobial action, dyes, or hydrophobic groups — several finishes of Section 2 are delivered by this one vehicle. It is wet-chemical, mill-compatible (pad–dry–cure logic), and its durability comes from the inorganic network's bond to the fibre rather than from a thick polymer film.

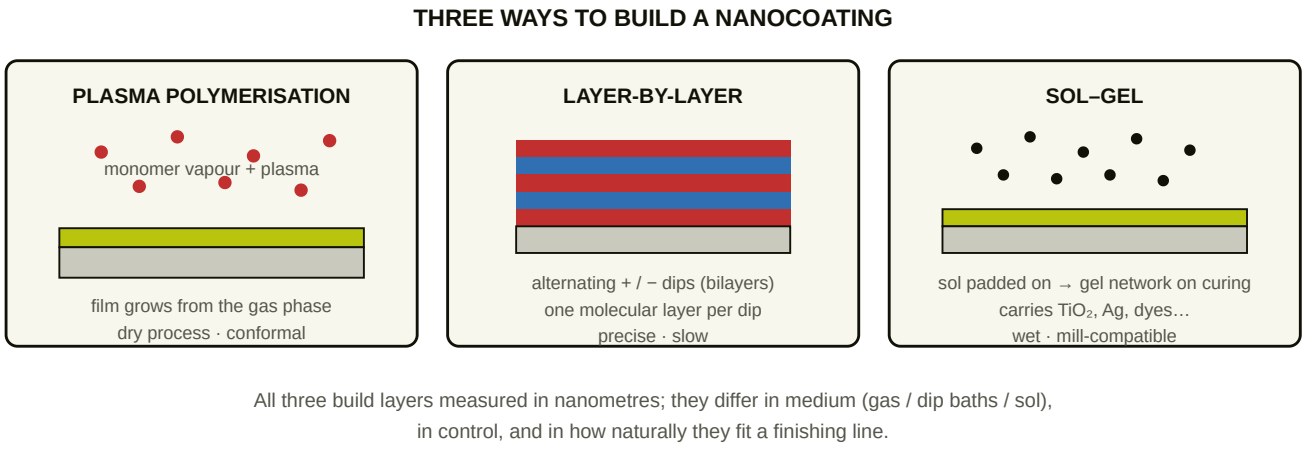


Figure 1 — Plasma polymerisation, layer-by-layer assembly and sol–gel coating compared schematically (grey band = fabric in each panel).

4. Worked example — dosing a sol–gel UV finish

WORKED EXAMPLE — ADD-ON FROM A PAD BATH

A sol–gel bath carries 20 g/L of TiO_2-bearing sol solids. The fabric is padded at 60% wet pick-up (0.60 L of liquor retained per kg of fabric).
Add-on = 20 g/L × 0.60 L/kg = **12 g/kg = 1.2% on weight of fabric**.
Interpretation: a complete UV-protective nano-layer is built from about a hundredth of the fabric's own mass — the efficiency argument of Section 1 in numbers. The specification is not complete until the UPF is measured on the cured *and washed* fabric: the dosage calculation plans the finish; the test certifies it.

5. Choosing among the routes

CRITERION	PLASMA POLYMERISATION	LAYER-BY-LAYER	SOL–GEL
Process medium	Dry (gas phase)	Wet (dip baths)	Wet (pad bath)
Thickness control	Good (time/power)	Finest (per bilayer)	Moderate (sol concentration, pick-up)
Mill compatibility	New equipment, batch tendency	Slow multi-dip; scale-up research	Fits pad–dry–cure lines directly
Best at	Ultra-thin conformal functional films	Designed multilayer architectures (FR stacks)	Delivering particles (TiO_2, Ag) in a bonded glassy host

Table 2 — Route selection. As in Topic 3, the decision is a screening: environment and line first, elegance last.

Glossary

Wet pick-up — liquor retained by fabric after padding, as a percentage of dry fabric mass. **Add-on** — finish solids deposited, as % of fabric mass. **UPF** — ultraviolet protection factor. **Bilayer** — one positive + one negative layer pair in LbL assembly. **Sol / gel** — a stable nanoparticle dispersion / the network it forms on drying and curing. **Conformal coating** — a film that follows the substrate's surface shape uniformly.

Key points

• Nano finishes trade on efficiency (grams per kilogram) and invisibility (handle unchanged) — and stand or fall on wash/abrasion durability.

• Self-cleaning is two mechanisms: photocatalytic TiO_2 (needs UV) and lotus-effect nano-roughness (needs its roughness to survive).
• Antimicrobial action is ion release; UV protection is nanoparticle absorption/scattering, rated as UPF on the washed fabric.
• Plasma polymerisation (dry, conformal), LbL (bilayer-precise, slow) and sol–gel (mill-ready particle host) are the three coating technologies — chosen by screening, like everything else in this course.
• Dosage arithmetic (bath g/L × pick-up) plans the finish; post-wash testing certifies it.

Practice

1. A pad bath contains 35 g/L sol solids and pick-up is 75%. Calculate the add-on in g/kg and % owf. The mill then halves the bath concentration to save cost — what add-on results, and what must be re-measured before the UPF label can stay?
2. A fabric transmits 4% of UV in a first test. Estimate its UPF, and state why the labelled UPF could still be lower after the standard test procedure.
3. Choose a coating route for (a) a flame-retardant multilayer stack whose architecture must be built layer by layer on cotton; (b) a continuous finishing line that must add TiO_2 UV protection to polyester dress fabric. Justify each briefly.

Answer key

1. Add-on = 35 × 0.75 = **26.25 g/kg ≈ 2.6% owf**. Halved bath: 17.5 × 0.75 ≈ **13.1 g/kg ≈ 1.3% owf**. The UPF must be re-measured on the cured and washed fabric at the new add-on — protection is not linear with dosage by assumption, and the label certifies a measurement, not a recipe.
2. UPF ≈ 1/0.04 = **25** as a first estimate. The labelled value can be lower because the standard procedure measures across the UV spectrum with weighting for skin sensitivity, on multiple specimens including after washing/stretching, and the rating is based on conservative statistics of those results — not on one transmission number.
3. (a) **Layer-by-layer assembly** — the requirement is an architecture of alternating functional layers (e.g. intumescent bilayers), which is exactly LbL's strength; its slowness is accepted for a high-value protective product. (b) **Sol–gel** — it runs on the existing pad–dry–cure logic, carries TiO_2 in a bonded silica network, and suits continuous dress-fabric production; plasma would demand new batch equipment for no functional gain here.